

CS:5980

Foundations of Embedded Systems

Safety Requirements

Part II

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A Brief Detour into Computational Complexity

Goal: Classify computational problems in terms of (roughly) how many basic operations it takes to solve the problem, as function of input size

Example 1: Finding maximum of a list of numbers

- Time complexity is linear: $O(n)$

Example 2: Sorting an array of numbers

- Algorithm (e.g., selection sort) with doubly-nested loop: $O(n^2)$
- More efficient algorithms (e.g., quicksort) possible: $O(n \log n)$

A Brief Detour into Computational Complexity

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Example 3: Boolean expression evaluation. Given

1. an expression e (with not/or/and as operations) over Boolean vars, and
2. an assignment a of 0/1 values to vars,

determine whether e evaluates to 1 or 0. Time complexity: $O(n)$

Example 4: Boolean satisfiability. Given a Boolean expression e , determine if there is an assignment a to vars that makes the expression evaluate to 1

- Naïve algorithm: Evaluate e on every possible assignment a
- Exponentially many choices for a : algorithm is $O(2^k)$, k = no. of vars

The Class P

Polynomial-time algorithm means a deterministic algorithm with time complexity such as $O(n)$, $O(n \log n)$, $O(n^2)$, $O(n^3)$, or $O(n^c)$, for some constant c

A problem is in **P** if there is a polynomial-time algorithm to solve it

Examples:

- Finding maximum
- Sorting
- Expression evaluation
- Finding shortest path in a graph

P is the class of *tractable* (i.e., efficiently solvable) problems

- Problem can be solved exactly
- Solution will scale reasonably well as input size grows
- In principle (asymptotically), $O(n)$ is better than $O(n^2)$

NP-Complete Problems

SAT: Given a Boolean expression e , check if there exists an assignment of 0/1 values to its variables for which e evaluates to 1

- No proof that SAT is in P (no known polynomial-time algorithm)
- No proof that SAT is not in P
- SAT is in $NP = \{ \text{problems where every candidate solution is checkable in polynomial time} \}$
- SAT is *NP-complete* (Cook, 1972)
- Hundreds of problems equivalent to SAT

Hamiltonian Path: Is there a path in a graph from source to destination that visits each vertex exactly once

Max Clique: Given a graph, find largest subset of vertices such that there is an edge between every pair of vertices in this set

Grand Challenge Open Problem: Is $P = NP$?

- If you find a polynomial-time algorithm for SAT, then $P = NP$, and many other problems will have polynomial-time algorithms
- If you prove SAT is not in P , then $P \neq NP$, and many other problems then provably don't have efficient algorithms

NP-Completeness Continued

Known algorithms for SAT are exponential-time in the worst-case, but

- Highly efficient implementations, SAT solvers, exist
- They can handle tens of millions of variables for real-world problems
- Many practical problems solved by encoding into SAT

Key feature of NP problems such as SAT: it suffices to find one satisfying assignment

This does not hold for all intractable problems

- **Boolean Validity:** Given a Boolean expression e , is it the case that e evaluates to 1 no matter what values we give to its variables

There are many complexity classes beyond NP: coNP , PSPACE , EXPTIME , ...

- Problems may require exponential-time (or more) to solve
- $P \subseteq NP \subseteq \text{PSPACE} \subseteq \text{EXPTIME} \subseteq \text{NEXPTIME} \subseteq \text{EXPSPACE}$
- $P \subsetneq \text{EXPTIME}$, $NP \subsetneq \text{NEXPTIME}$ and $\text{PSPACE} \subsetneq \text{EXPSPACE}$

(Un)Decidability

Some general problems cannot be solved by a computer at all!

Fundamental Theorem of CS (Turing, 1936):

- The *Halting problem* for Turing machines is **undecidable**:
There is no program that takes as its input an **arbitrary** program C and an **arbitrary** input x , and determines if C terminates on x

Intuition: If a program could analyze other programs exactly, then it can analyze itself, and this suffices to set up a logical contradiction!

A surprisingly undecidable problem: Does a given a polynomial (e.g., $x^3 + 2xy^2 - 15xy + 156$) have integer roots?

Decidable Problem: There exists a program, or Turing machine, that solves all instances of the problem correctly (i.e., gives the right answer and stops)

- Includes problems in P as well as intractable classes such as NP , $Exptime$, etc.

Back to Invariant Verification Problem



Theorem: The invariant verification problem is **undecidable**

Proof idea: undecidable problems for Turing machines can be recast as invariant verification problems for transition systems with integer state variables

Finite-State Invariant Verification Problem

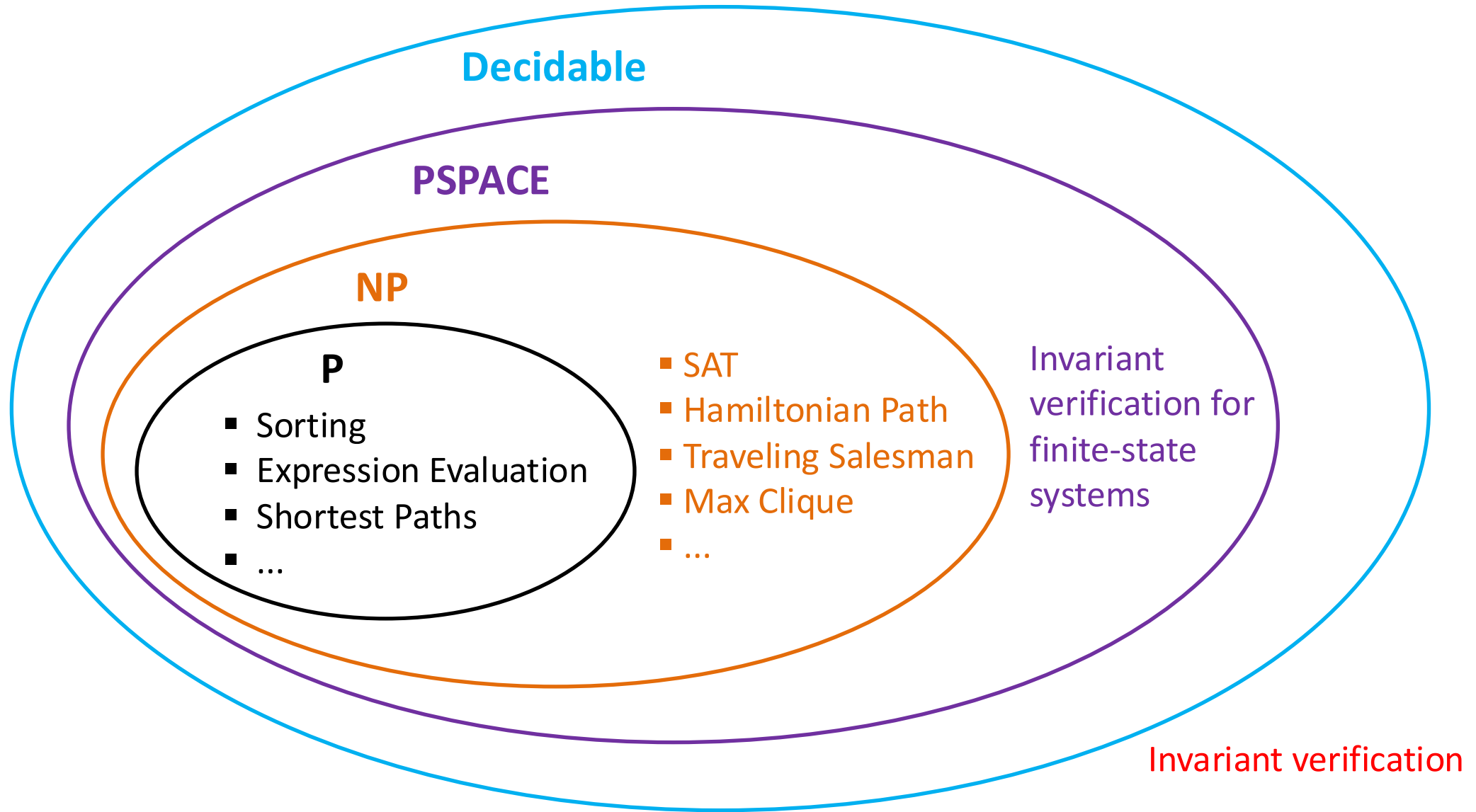


Theorem: The invariant verification problem for finite-state systems is **decidable**

Proof sketch: If T has k Boolean state vars, then total number of states is 2^k .

Verifier can systematically search through all possible states.

Complexity is **exponential**. More precisely, it is **PSPACE-complete**, and so harder in practice than **NP-complete** problems such as SAT



Credits

Notes based on Chapter 3 of

Principles of Cyber-Physical Systems

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